

DEEP LEARNING

Complete Reference Notes for Real-World Project Work



Compiled from personal XMind mind map — organized as a quick-recall guide covering core concepts, network types, libraries and real-world applications.

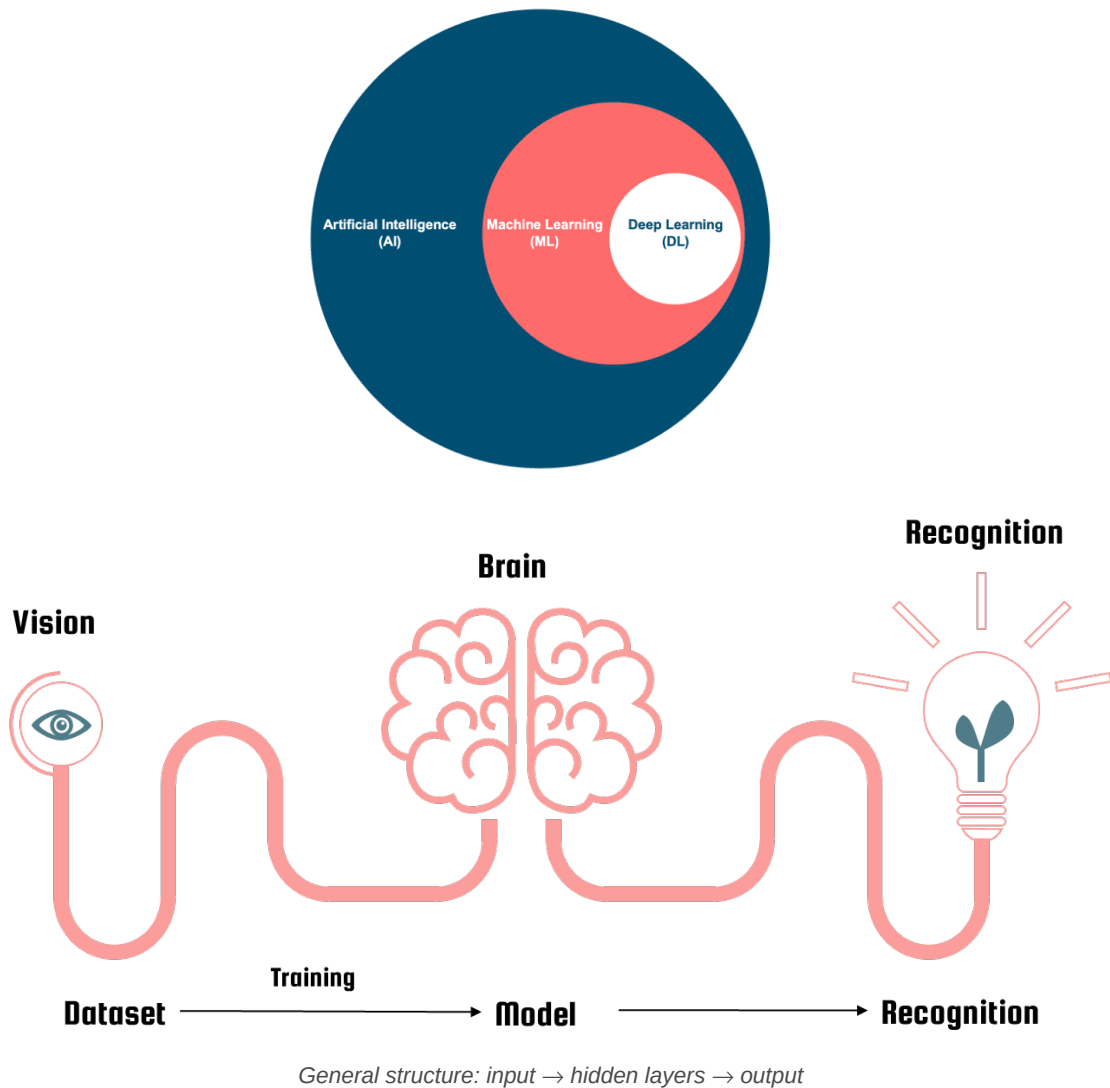
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What is Deep Learning

Definition

Deep learning is a subset of machine learning where artificial neural networks, inspired by the human brain, are trained to recognize patterns and make decisions.



Why Deep Learning (over classic ML)

Automatic Feature Extraction: Traditional Machine Learning needs a human to manually engineer features. Deep Learning automatically learns the important features directly from raw data (images, text, audio) through its layers — this is the single biggest practical advantage when you're short on domain expertise for feature engineering.

◆ Machine Learning (ML)

- In ML, you must manually create features.
- Example: For face recognition,
 - You manually extract eyes distance, nose width, edge shapes, etc.
 - Then give these features to an ML model (like SVM).

👉 ML depends on human-designed features.

◆ Deep Learning (DL)

- Deep learning automatically extracts features from raw data.
- Example: For face recognition,
 - You give the model raw images.
 - The neural network automatically learns:
 - edges → shapes → eyes/nose → full face patterns.

👉 DL discovers features by itself without manual work.

What Deep Learning Requires

- **Massive amount of data** — deep networks need large labelled/unlabelled datasets to learn reliable patterns.
- **High computational power** — training involves millions of matrix operations, so it needs strong hardware.

Why GPU over CPU

CPU Central Processing Unit — few powerful cores, great for sequential/general-purpose tasks.

GPU Graphical Processing Unit — thousands of smaller cores that accelerate complex, parallel computations (originally for gaming graphics). This parallelism massively speeds up the matrix math deep learning relies on, making GPUs essential for practical training times.

■ *Project tip: for any real training job beyond toy datasets, use a GPU (local, Colab, or cloud) — CPU training will be a major bottleneck.*

Core Building Blocks

2.1 Neuron

The basic unit of a neural network. A neuron takes input(s), applies a mathematical function to them, and produces an output.

Human Brain Neuron

Imagine you see a dog.

Your brain receives inputs:

- Eyes → shape
- Ears → sound
- Nose → smell

Your brain neuron **combines all these signals** and decides:

👉 "Yes, this is a dog."

Artificial Neuron (in Deep Learning)

Inputs:

- Pixel ① = 0.8
- Pixel ② = 0.3
- Pixel ③ = 0.6

The artificial neuron **combines these numbers using a math formula**:

$$\text{output} = (x_1w_1 + x_2w_2 + x_3w_3)$$

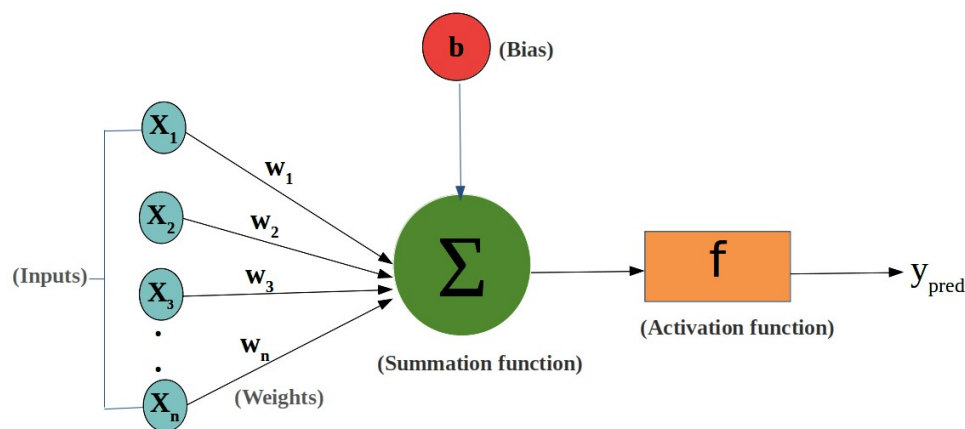
If the result is high, the neuron decides:

👉 "This looks like a dog."

2.2 Layer

A layer is made up of nodes (neurons) and weights. Every network has three kinds of layers:

- **Input Layer** — receives the raw data.
- **Hidden Layer(s)** — process and transform data between input and output.
- **Output Layer** — produces the final prediction/result.



Basic neural network structure

Layer types you'll actually use

- **Fully Connected (Dense) Layer** — every neuron in this layer connects to every neuron in the previous layer. Used mostly near the end of a network for final classification/regression.
- **Convolution Layer** — scans small parts of an image with a filter to detect patterns like edges, corners, and shapes (e.g. scanning an elephant image to detect trunk, tusks, ear edges). Core of CNNs, used for image data.
- **Pooling Layer** — simplifies data by keeping only the most important information, shrinking the size (like summarizing a long story into key points). Reduces computation and overfitting.

2.3 Weights & Bias

Weights decide how important each input is. They multiply with the inputs to increase or decrease their effect, guiding the network toward accurate predictions.

Bias lets the neuron make useful decisions even when inputs are zero — if all inputs = 0, the bias still gives the neuron a starting value to produce meaningful output.

Imagine you are judging a student's performance.

You consider 3 things:

- Homework
- Exams
- Attendance

But you feel **Exams are more important**, so you give it more weight.

Weights Example

- Homework weight = 0.2
- Exams weight = 0.7
- Attendance weight = 0.1

If the student scores:

- Homework = 80
- Exams = 90
- Attendance = 100

Neuron calculates:

$$\text{Score} = (80 \times 0.2) + (90 \times 0.7) + (100 \times 0.1)$$

Weights decide which input matters more.

👉 **Meaning:**
Weights tell the model which input is more important.

2.4 Activation Function

A mathematical formula that decides whether a neuron should “activate” (produce output) or not — this is what lets a network learn non-linear, real-world patterns instead of just straight lines.

Function	What it does
Step	Output is 1 if $X \geq 0$, else 0 — a hard cutoff.
Sigmoid	Squashes any number into a smooth value between 0 and 1 (like a probability).
Softmax	Converts raw output scores into a probability distribution across classes — used for multi-class classification output.
Tanh	Squashes values into a range of -1 to +1, giving both negative and positive outputs.
ReLU	Keeps positive values as-is and turns negative values to zero. Default choice for most hidden layers today.

■ *Project tip: use ReLU in hidden layers by default; use Sigmoid for binary output, Softmax for multi-class output.*

2.5 Hyperparameters

Variables that determine the network structure and how it's trained — you set these before training, the network doesn't learn them on its own.

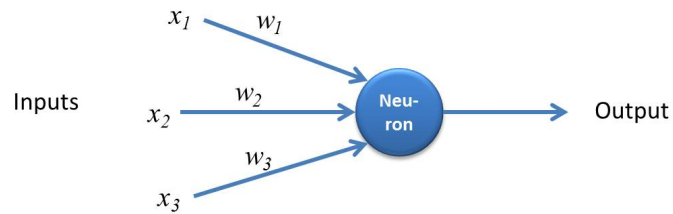
- **Dropout** — randomly turns off neurons during training to prevent overfitting.
- **Activation Function** — choice of ReLU / Sigmoid / Softmax / Tanh per layer.
- **Learning Rate** — controls how big a step the model takes while learning; too high = unstable, too low = slow.
- **Number of Epochs** — how many times the model passes through the entire training dataset.
- **Batch Size** — how many samples are processed before the model's weights are updated.
- **Number of Neurons** — how wide each layer is, affecting model capacity.

■ *Project tip: these are exactly the knobs you'll tune during hyperparameter tuning / grid search on a real project.*

3 Types of Neural Networks

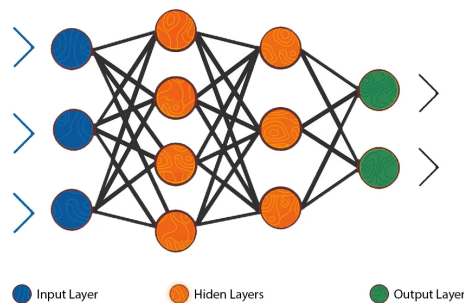
1. Perceptron

The simplest form of neural network — a single-layer network with one neuron performing a binary decision.



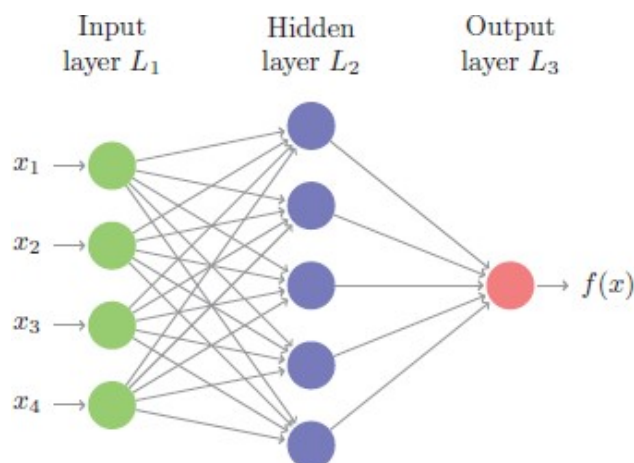
2. Multilayer Perceptron (MLP)

Its decision function is a step function and the output is binary. Stacks multiple layers of perceptrons to handle more complex problems.



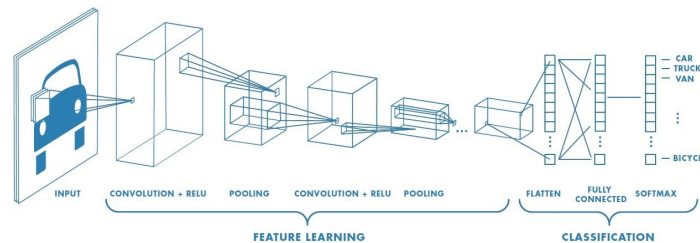
3. Feed Forward Neural Network (FFNN)

The simplest form of neural network where input data travels in one direction only, passing through the nodes without looping back.



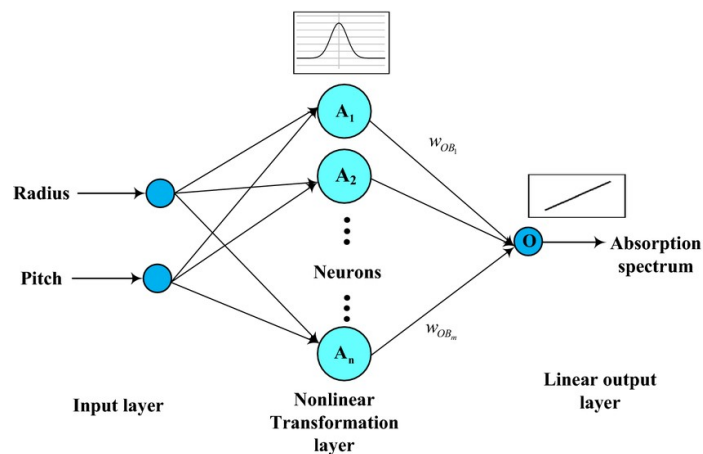
4. Convolutional Neural Network (CNN)

Contains a three-dimensional arrangement of neurons; each neuron processes information from a small part of the visual field. Best for images/video.



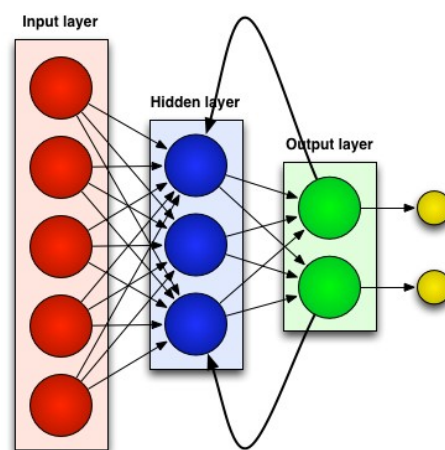
5. Radial Basis Function Neural Network (RBFNN)

Transforms the input signal into another form that can then be fed into the network to achieve linear separability. Has exactly one hidden layer.



6. Recurrent Neural Network (RNN)

Remembers information from previous inputs to influence the current output — ideal for sequences like text, speech, and time series.



■ *Quick pick guide: images/video → CNN. Sequences/text/time-series → RNN (or its variants LSTM/GRU). Simple tabular data → MLP/FFNN.*

Popular Deep Learning Libraries

These are the standard tools used to actually build the concepts above in real projects.



TensorFlow

Google's end-to-end open-source platform for building and deploying ML/DL models at scale, including production and mobile deployment.



Keras

A high-level, user-friendly API (now part of TensorFlow) for quickly building and training neural networks with minimal code.



PyTorch

Facebook's (Meta's) flexible, Python-first deep learning framework — the most popular choice in research and increasingly in production.



Caffe

A deep learning framework focused on speed, historically popular for computer vision applications.



MXNet

A scalable, efficient deep learning framework supporting multiple languages, backed by Apache.



NumPy & SciPy

Foundational Python libraries for numerical computing and scientific operations — used under the hood for array/matrix math in almost every DL pipeline.

Application	What it does
Fruits & Vegetable Classification (Segregator)	Identify and separate fruits/vegetables based on images to sort them faster.
Language Translation (NLP)	Convert one language to another (e.g., English → Tamil) just by learning from huge text datasets.
Sentiment Analysis (NLP)	Understand emotions in text (positive/negative) from reviews, comments, or tweets.
Autonomous Vehicles	Cars see roads, detect objects, and make driving decisions automatically.
Medical Field	Identify diseases from X-rays and MRI scans, or predict health risks from patient data.
Agriculture	Detect crop diseases, count plants, predict yield, and guide smart farming.
Object Recognition	Detect and label objects in images (car, person, dog, bottle, etc.).
Face Recognition	Identify or verify a person's face for security, attendance, or mobile unlocking.

One-Page Cheat Sheet

Use this as a fast lookup while working on a project.

Topic	Key takeaway
Deep Learning	Subset of ML using brain-inspired neural networks to find patterns automatically.
Why use it	No manual feature engineering — the network learns features itself, given enough data.
Needs	Large datasets + strong compute (prefer GPU over CPU for training).
Neuron	Takes input → applies a function → produces output.
Layer	Input → Hidden → Output. Dense = fully connected; Conv = pattern scanning; Pooling = downsampling.
Weights & Bias	Weights = importance of each input; Bias = baseline output even when input is 0.
Activation	ReLU (hidden layers, default) · Sigmoid (binary output) · Softmax (multi-class output) · Tanh (-1 to 1).
Hyperparameters	Learning rate, epochs, batch size, dropout, number of neurons — tune these during training.
Network to pick	Images → CNN · Sequences/text → RNN · Tabular/simple → MLP/FFNN.
Libraries	TensorFlow, Keras, PyTorch, Caffe, MXNet + NumPy/SciPy for the math.

End of notes — compiled from your Deep Learning XMind mind map.